Elective - I: Cryptography & Network Security

Course Code	Course Name	Category	Hours / Week	Credits
24BCA5BE	Cryptography & Network Security	Elective - I	5	3

Course Objectives

The course intends to cover

- Basic concepts of Cryptography.
- Digital certificates and Public Key Infrastructure (PKI) protocols.
- Authentication techniques.

Course Learning Outcomes

On the successful completion of the course, students will be able to

CLO	CLO Statements	Knowledge Level
CLO1	Understand the basic concepts of Cryptography.	K1,K2
CLO2	Interpret symmetric and asymmetric cryptography Algorithms.	K2
CLO3	Apply Internet security protocols for communication.	K3
CLO4	Understand authentication mechanisms and cryptography.	K2
CLO5	Analyze firewalls, VPNs and network security mechanisms.	K4
K1-Remember; **K2**-Understand; **K3**-Apply;**K4**-Analyze;		

CLO – PLO Mapping

CLOs/PLOs	PLO1	PLO2	PLO3	PLO4	PLO5
CLO1	3	3	3	3	3
CLO2	2	3	3	3	2
CLO3	3	3	3	3	1
CLO4	3	2	3	3	1
CLO5	3	2	2	3	1
3 - Substantial (high)		**2 – Moderate(medium)**		**1 - Slight (low)**	

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Unit	Content	No. of Hours
I	**Attacks on Computers and ComputerSecurity:**Introduction - Need for Security - Security approaches - Principles of Security - Types of attacks. **Cryptography Techniques:** Basic terms - Plain text and Cipher text - Substitution techniques - Transposition techniques - Encryption and decryption - Symmetric and Asymmetric key cryptography.Applications of AI in Cryptography.	15
II	**Symmetric Key Algorithms and AES :** Introduction - Algorithm Types and Modes - An overview of Symmetric key Cryptography - Data Encryption Standard (DES) - Blowfish - Advanced Encryption Standard (AES). **Asymmetric Key Algorithms**: Introduction - Brief History of Asymmetric Key cryptography - An Overview of Asymmetric Cryptography - The RSA algorithm - Symmetric and Symmetric Cryptography together - Digital Signatures.	15
III	**Public Key Infrastructure (PKI):** Introduction - Private Key Management - The PKIX Model - Public key Cryptography standards - XML, PKI and Security -**Internet Security Protocols :** Introduction - Basic Concepts - Secure Socket Layer - (SSL) - Transport Layer Security(TLS) - 3-D secure Protocol - Electronic Money - Email Security.	15
IV	**User Authentication and Kerberos:** Introduction - Authentication basics - Passwords - Authentication Tokens - Biometric Authentication - Kerberos - Key Distribution Centre - Security Handshake Pitfalls. **Cryptography in JAVA, .NET:**Introduction - Cryptographic Solution using JAVA - Cryptographic Solutions using Microsoft .NET Framework – Cryptographic Toolkits - Security and Operating Systems - Database Security.	15
V	**Network Security Firewalls and Virtual Private Networks (VPN) :** Introduction - Brief introduction to TCP/IP - Fire walls-Types of Firewall Configuration - Limitation of Firewall - IP security - IPSec Overview-IPSec Key Management - Virtual Private networks (VPN) - Intrusion.	15
	Total Hours	**75**

Text Books	
1.	AtulKahate (2019), Cryptograpy and Network Security, 4th Edition, Tata McGraw-Hill Publishing.
2.	William Stallings (2023), Cryptography and Network Security Principles and Practices, 8th edition, PHI Education Asia.
Reference Books	
1.	Andrew S. Tanenbaum(2021),Computer Networks, 6th Edition, PHI.
2.	Bernard L Menezes,andRavinder Kumar (2018) "Cryptography,Network Security and Cyber Laws",Cengage Learning India Pvt Limited.
3.	Behrouz A. Forouzan (2015),Cryptography and Network Security,TMH.
Web Resources	
1.	https://nptel.ac.in/courses/106/105/106105167/**(Swayam / NPTEL)**
Other Online Resources	
2.	https://www.tutorialspoint.com/computer networks/index.htm
3.	https://www.javatpoint.com/computer networks-tutorial
4.	https://drive.google.com/file/d/1v0lj53dG3V-rVt73uxISN38vmizRbOxD/view?usp=sharing